Layered GPS Spoofing Detection for Small Drones: Simulation-Based Model Selection and Evaluation

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***Abstract*—A spoofer can make a drone's GPS receiver report a false position by broadcasting stronger fake satellite signals. We propose a detector of ten checks whose answers are combined into one decision, and we evaluate its parts separately. On public flight data, we repeat a published classifier test and find that its high scores come mainly from a receiver reading that records the attack. A GPS-only motion check fails on real recorded flights. In a physics-based simulator, we compare 13 learning algorithms under a rule fixed before the results were seen. The selected model, a one-class SVM trained on clean flights, caught 96.0% of attacks on a separate simulated test set, with 0.77 false alarms per hour (95% range 0.44 to 1.25) and a median delay of 0.99 s. Its weakest case is consistent takeover, caught 73% of the time. Without the signal-strength clue, detection on validation data falls to 92% at 0.96 false alarms per hour, so the result depends on the simulator's model of attack power. No hardware or flight results are reported.**

***Index Terms*—GNSS spoofing, unmanned aerial vehicles, sensor fusion, anomaly detection, model selection, simulation.**

I INTRODUCTION

Small drones rely on GNSS signals for position. These signals are unauthenticated, so a software-defined radio can send counterfeit signals that override the real ones [1]. A drone that trusts the result can be led off its route without any visible warning [2, 3].

Published detectors differ in cost, latency and assumptions. Several rely on a single kind of evidence, and some report high accuracy on random splits that can place samples from one flight in both training and test sets [3]. We ask a narrower question: given a set of simple checks, which learning algorithm should combine them, and how much does each part of the detector contribute?

Our contributions are: (1) a repeat of a published classifier test on public data, with a shortcut identified; (2) a test of a GPS-only motion check on real recorded flights; (3) a simulator of drone flights with clean and six attack types; (4) a comparison of 13 algorithms under a pre-set selection rule, with one selected model tested once on new data; and (5) a simulated clock check. Section II places these against four published methods.

II RELATED WORK

Nayfeh et al. [2] train classifiers on receiver features for flights of 20 to 70 m squares and report 92.36% three-class accuracy with a decision tree. We kept their feature family and their held-out-square split. Our repeat matches their best setting within 0.5 points, but a single receiver reading gives 99.3% binary accuracy on its own (Fig. 2, 3).

Gormus et al. [3] report XGBoost at 96.40% accuracy, while their own table shows a random forest at 97.72%. Their dataset is available only on request, so we did not run the method. We took the idea of edge deployment and their feature set.

Xie et al. [4] propose a two-layer detector whose first layer compares position change with reported velocity over windows. We implemented that first layer and tested it on real flights, where it caught no attacks (Table 4). We took its residual definition and threshold rule. The second layer needs an inertial reference, which the public data lacks.

Spanghero and Papadimitratos [5, 6] use a clock filter to detect time attacks, on a hardware testbed. We implemented the clock-innovation test in simulation only and report it as such.

III PROPOSED DETECTOR

The detector has ten checks (Table 1). Their answers are combined by a vote, and the simulation study in this paper evaluates the learned combination of 19 features described in Section V.

TABLE 1THE TEN CHECKS AND WHERE EACH HAS BEEN TESTED

#	Check	Tested so far
1	Clock jump or drift	Simulation
2	Signal strength and noise	Public data; simulation
3	Satellite layout (DOP, count)	Public data; simulation
4	Agreement between satellite systems	Not tested (needs second solution)
5	Position against speed	Public data; simulation
6	Altitude and acceleration against sensors	Public data (partial); simulation
7	Raw carrier signal	Not tested (needs raw data)
8	Unusual-behaviour score	Public data
9	Clock stability over long periods	Not tested
10	Learned model	Public data; simulation

IV SYSTEM AND THREAT MODEL

A drone reports GNSS position and velocity, with signal strength and satellite geometry, and carries an IMU and barometer. The attacker transmits counterfeit L1 signals and may choose a false path, a step, or a ramp. The attacker's power is 0 to 6 dB above the authentic signal in our main simulation, and 0

to 1 dB in a weak-power test. The attacker does not control the IMU or barometer and does not jam the authentic signal.

V METHOD

A *Simulated data*

The simulator produces clean flights and seven attack types: static, step, ramp, step then ramp, ramp then step, consistent takeover, and dynamic takeover. In a consistent takeover, the reported velocity matches the spoofed path. In other types, Doppler velocity still reflects the true motion. Every attacked session raises signal strength on all satellites by the spoofer's power. The clock is not changed by the spoofer, and there is no multipath. Features are causal, computed over trailing windows. The data are split into training, calibration and validation sets for selection, plus a separate test set with a new random seed, opened once.

B *Algorithms and settings*

We compared 13 algorithms: logistic regression, decision tree, random forest, extra trees, histogram gradient boosting, XGBoost, LightGBM, linear SVM, RBF SVM, a two-layer neural network, k-nearest neighbours, Isolation Forest, and a one-class SVM. The last two are trained on clean flights only, because real attacks may be rare or unlike the training attacks. Each supervised model was tuned on a small grid, fixed before the runs, using validation data only.

C *Selection rule*

The rule was written before the runs. (1) A model is eligible if it has at most 1 false alarm per hour at the calibrated threshold and a non-zero detection rate with signal strength removed. (2) Eligible models are ranked by detection rate on validation data. (3) Robustness, the mean of weak-power and unseen-attack detection, is also considered. (4) Models within 0.05 of each other are tied, and the tie goes to the lowest median delay. Alarms need 10 consecutive rows above threshold.

VI RESULTS

A *Model comparison on validation data*

Table 2 lists the models that passed eligibility, with the one-class SVM selected. Twelve of 13 models passed the eligibility check. Nine were tied on detection and robustness, and the tie-break on delay selected the one-class SVM.

TABLE 2 VALIDATION RESULTS FOR SELECTED MODELS (5-SEED MEAN FOR DETECTION).

Model	Detection	FA/h	Delay (s)
One-class SVM (selected)	0.960	0.90	0.99
LightGBM	0.984	0.06	1.45
XGBoost	0.976	0.00	1.57
MLP	0.976	0.18	1.30
Logistic regression	0.960	0.54	1.23
Random forest	0.937	0.12	2.21
Extra trees	0.857	0.36	2.15
RBF SVM (not eligible)	0.381	1.38	5.29
Isolation Forest	0.032	0.54	1.51
Decision tree	0.000	0.00	–

The selection shows a trade-off that the rule does not settle. LightGBM has the highest detection and lower false alarms, but a delay about 0.45 s longer. The one-class SVM has the highest false-alarm rate of the eligible models. Fig. 1 shows the trade-off. The decision tree and Isolation Forest also pass the eligibility step, although they detect almost nothing. This shows that the rule needs a minimum detection rate. This did not change the choice.

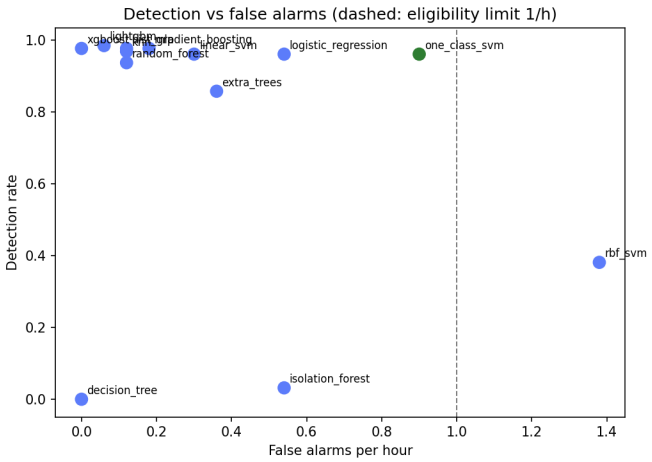


FIGURE 1. Detection rate against false alarms per hour for the 13 candidates on validation data.

B *Final test on new data*

The one-class SVM was locked before the test set was opened. On 1,500 clean and 420 attacked sessions, it caught 403 of 420 attacks (0.960; 95% range 0.936 to 0.975). It raised 0.77 false alarms per hour, with 16 events in 20.8 clean hours (95% range 0.44 to 1.25). Median delay was 0.99 s, and the 90th percentile was 9.3 s. Decision time was 0.027 ms.

TABLE 3FINAL TEST BY ATTACK TYPE (60 ATTACKED SESSIONS PER TYPE).

Attack	Detected	95% range
Static	1.00	0.94 to 1.00
Step	1.00	0.94 to 1.00
Ramp	0.98	0.91 to 1.00
Step then ramp	1.00	0.94 to 1.00
Ramp then step	1.00	0.94 to 1.00
Consistent takeover	0.73	0.61 to 0.83
Dynamic takeover	1.00	0.94 to 1.00

The false-alarm figure has a range whose upper end (1.25 per hour) is above the 1 per hour target, so the target is met on the point estimate but not with certainty.

C *Signal strength and weak power*

Removing the signal-strength and DOP fields reduces validation detection to 0.92 at 0.96 false alarms per hour. Because every simulated attack raises signal strength, this result depends on that modelling choice.

D *Earlier results*

On the public Nayfeh data, our repeat is within 0.5 points of the published best setting. The jamming indicator alone gives 99.3% binary accuracy (Fig. 3). A GPS-only motion check on real flights caught none of the attacks (Table 4). On the public data, a vote over six checks caught all attacks, with false-alarm rates of 0.000 at 20 and 35 m, 0.010 at 50 m, and 0.221 at 70 m (Fig. 4). In simulation, the clock check caught every tested time attack in 2 to 6 s, with no false alarm in 30 simulated hours (Fig. 5). This assumes the filter matches the simulated oscillator.

TABLE 4GPS-ONLY MOTION CHECK ON REAL RECORDED FLIGHTS.

Flight type	Attacks caught	False alarms
Hovering (static)	0 of 31 windows	none
Moving (dynamic)	0 of 50 windows	none

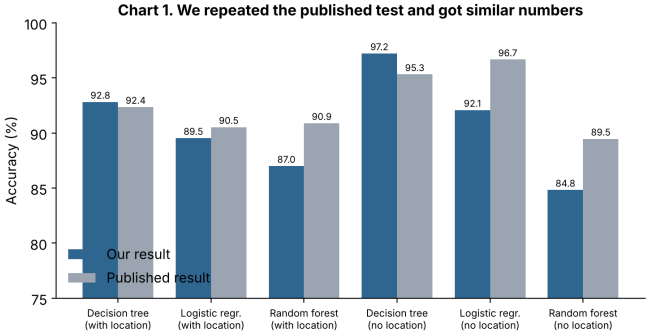


FIGURE 2. Repeat of the published test compared with the paper.

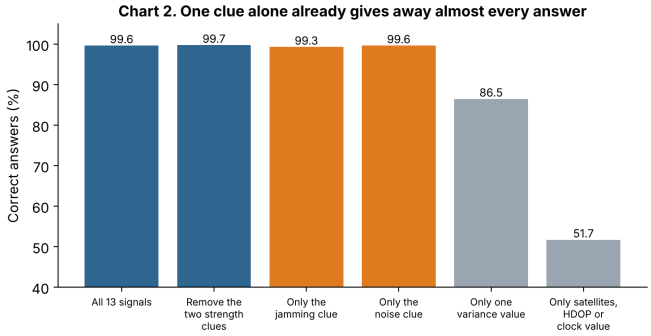


FIGURE 3. Share of correct answers from different sets of receiver readings.

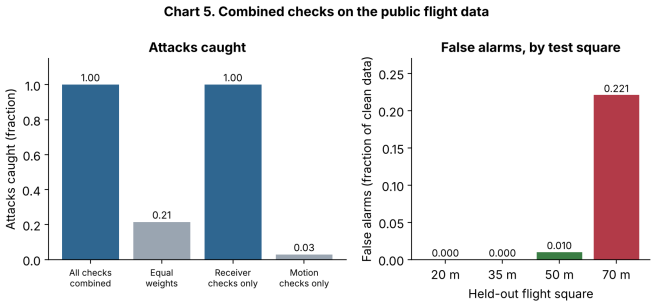


FIGURE 4. Public-data vote: attacks caught (left) and false alarms by held-out square (right).

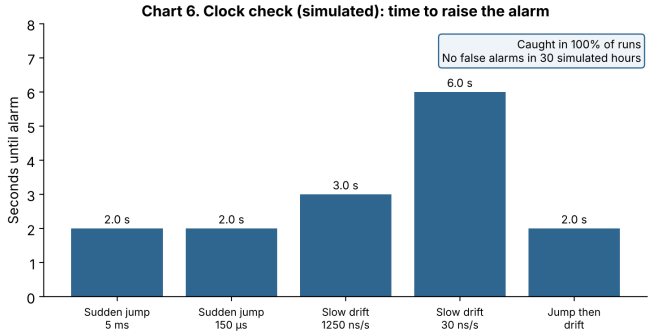


FIGURE 5. Simulated clock check: time from attack start to alarm.

VII DISCUSSION AND LIMITATIONS

Our results are simulation-based, except for the public-data and real-flight tests. The simulator assumes that spoofers raise signal strength and that the clock is unchanged by the attack. Both assumptions favour the signal-strength check. The consistent takeover, which keeps position and velocity consistent, is the weakest case. It is the attack a GPS-only check cannot see, and it needs an independent sensor. The public-data vote's 70 m false-alarm rate is unexplained. We did not test on hardware, and the sim-to-real gap is unmeasured.

The selection rule has a gap: it lacks a minimum detection rate, and the choice depends on how delay and false alarms are weighed. The model was selected on the same simulator that

generated the test data, with a new seed. This shows generalisation within the simulator, not to real receivers.

VIII CONCLUSION

We proposed a ten-check GPS spoofing detector and evaluated its parts. On a separate simulated test set, a one-class SVM trained on clean flights detected 96% of attacks with 0.77 false alarms per hour, and failed mainly on consistent takeover. The result depends on the simulator's attack model. Next steps are to repeat the final check on public data and on the first bench recordings, add a longer-window feature for consistent takeover, and fix the selection rule's detection gap.

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